

Stablecoin Yield

The price of yield: persistence, mechanisms and TVL response in stablecoin DeFi

CENTRAL CLAIM

Stablecoin APY becomes interpretable only when duration, TVL, mechanism and peg context are shown together.

Yield as a regime, not a ranking.

Simone Ragusini · 945119

s.ragusini@campus.unimib.it

Educational analysis only. No financial advice, no pool ranking.

Evidence snapshot

A compact view of the analytical scale

250

POOLS

137k

POOL-DAYS

4.3%

POOL-DAY MEDIAN APY

2

MEDIAN EPISODE
DURATION

Yield is a regime, not a rank

APY = quoted annualized yield. TVL = total value locked, measured in USD.

ONE ANALYTICAL GRAIN

POOL × DAY

Each observation connects yield level, capacity, mechanism and market context at a defensible temporal resolution.

Boundary: APY is quoted annualized APY. TVL change is an observed proxy, not direct wallet-level flow.

01

How long do episodes at or above 10% APY last?

02

Which mechanisms explain headline APY?

03

How stable are top-yield rankings?

04

What changes during peg stress?

05

Which pools clear all three thresholds?

The sample is broad, but deliberately selected

2022-02-11 to 2026-07-08 | 137,095 pool-days | unbalanced panel

Selected pools by analytical type



SELECTION LOGIC

Eligibility	Stablecoin pool, TVL $\geq$ USD 1M, at least 180 provider history observations.
Cap of 250	70% highest TVL; remaining capacity highest current APY. No random sampling claim.
Entry and exit	Observed pool-specific dates only. No synthetic backfill or balanced-panel imputation.

Data quality 26 missing TVL rows | 32 APY > 1,000% flagged, not hidden

Four definitions make the analysis auditable

Each result has an explicit threshold, comparison window and interpretation boundary.

01

High-yield episode

APY at or above 10%; contiguous observed run; one missing calendar day tolerated; active runs right-censored.

Kaplan-Meier retains censored episodes.

02

Ranking churn

1 - intersection share of top-k sets at t and t+h; k = 10 or 20; h = 1, 7 or 30 days.

Stability diagnostic, not a leaderboard.

03

APY-jump event

APY increase ≥ 5 pp on consecutive calendar days and APY $\geq 10\%$. First 3 events per pool. Window: -7 to +30 days.

450 events at day 0; TVL remains observational.

04

Joint threshold screen

At or above the sample medians for pool-level APY, persistence and TVL simultaneously.

Descriptive three-threshold screen; not a Pareto frontier.

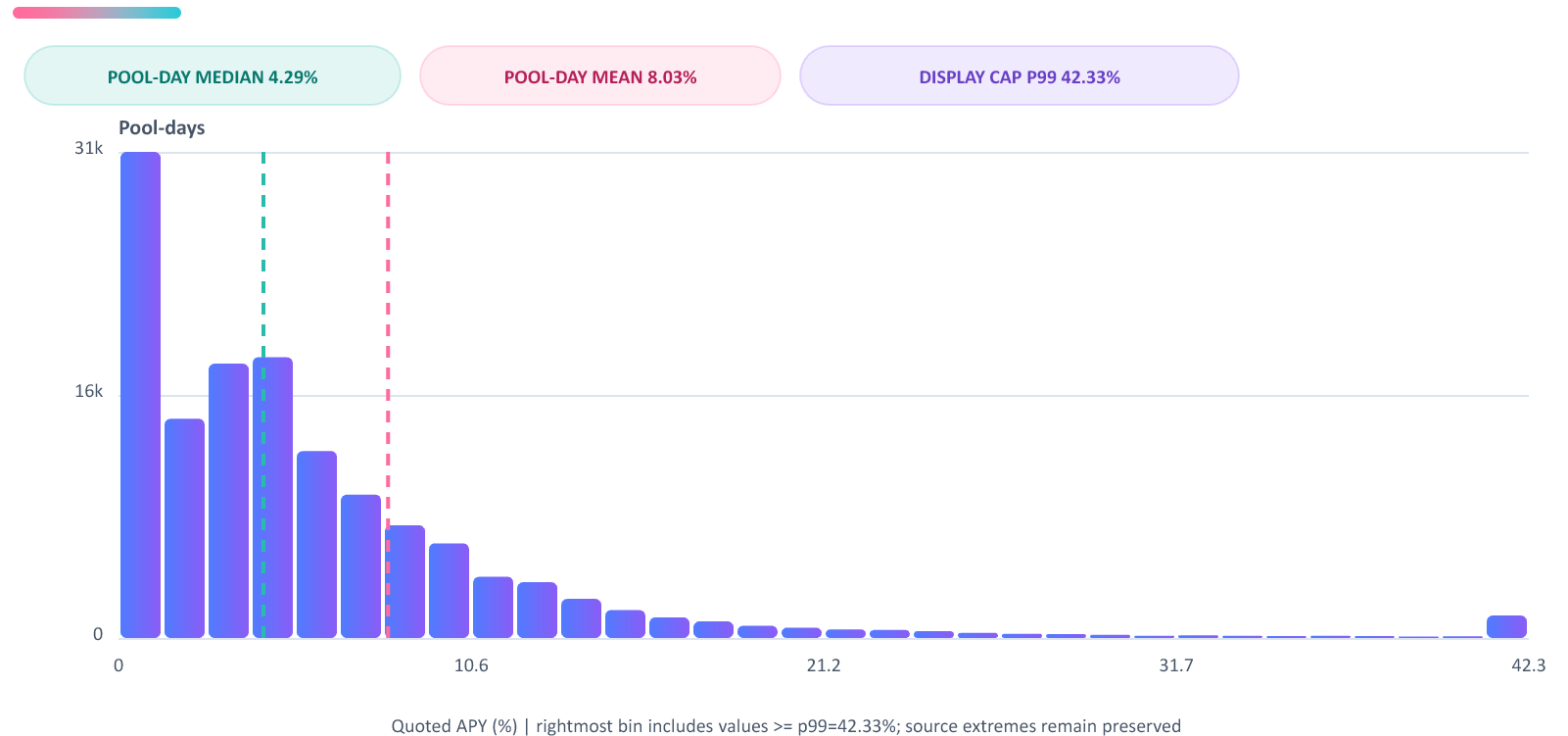
Headline APY needs robust summaries

Why are headline APY values hard to compare?

WHAT THE CHART SAYS

Pool-day median APY is 4.29% while the pool-day mean is 8.03%.

- Pool-day median APY: 4.29%
- Pool-day mean APY: 8.03%
- Extreme observations are flagged and preserved

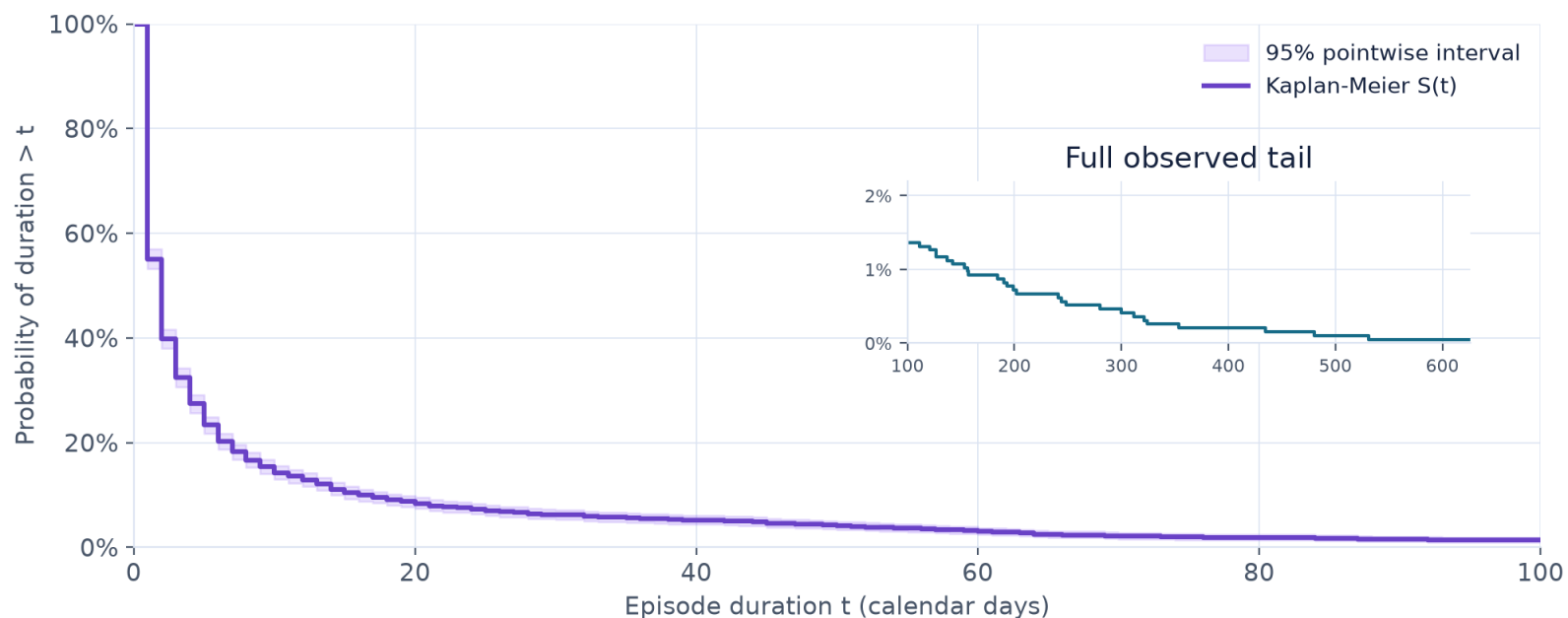


Sources: DeFiLlama, CoinGecko checks and protocol documentation. Quoted APY differs from realized return.

High yield usually behaves like a short regime

How long do episodes at or above 10% APY last?

2,668 episodes | $S(30) = 6.22\%$ | 95% CI: 5.34-7.19%



WHAT THE CHART SAYS

The median high-yield episode lasts 2 days; only 6.2% remain after 30 days.

- 2,668 episodes at or above 10% APY
- Median duration: 2 days
- 6.2% survive beyond 30 days

Sources: DeFiLlama, CoinGecko checks and protocol documentation. Quoted APY differs from realized return.

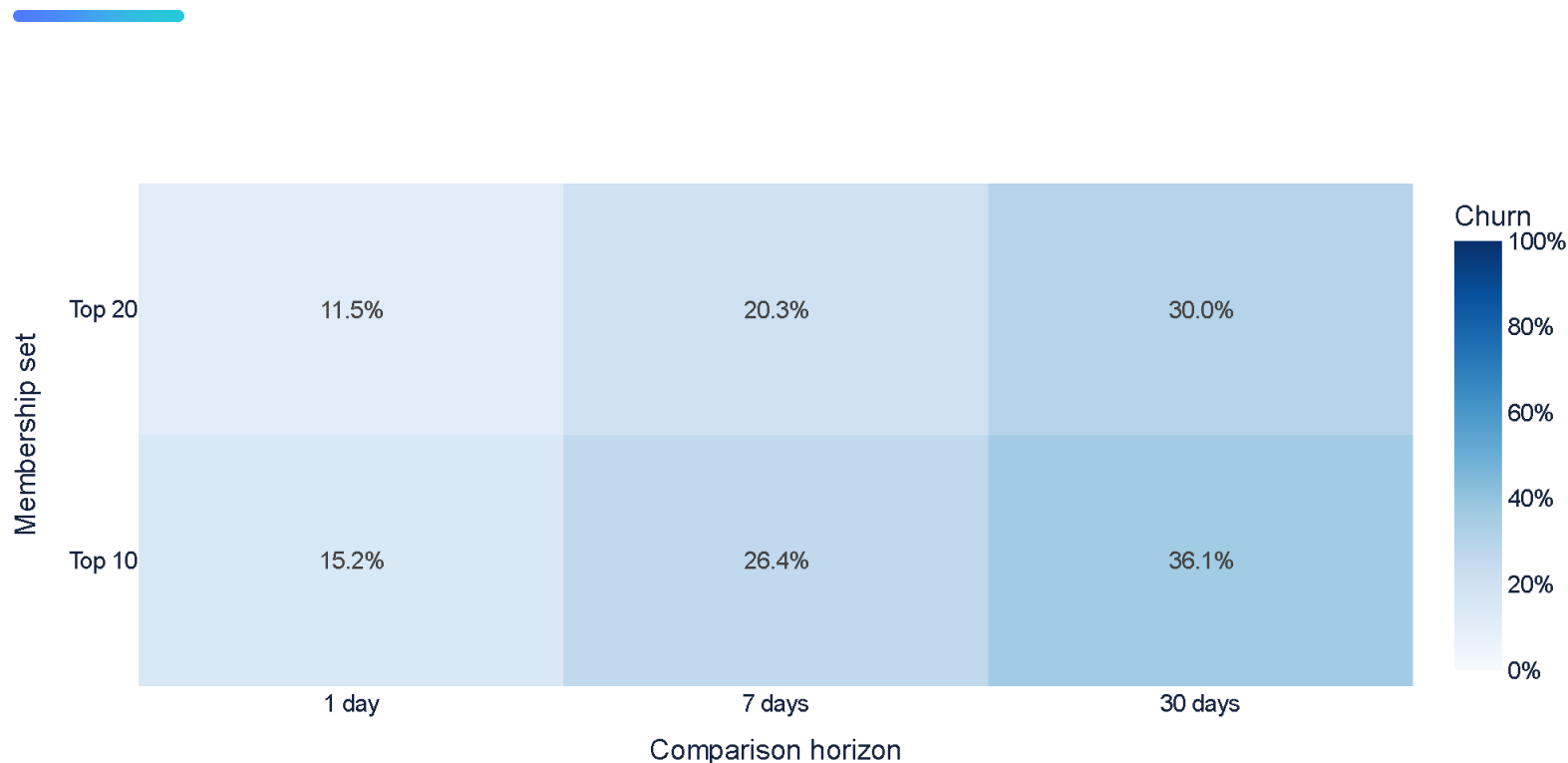
Top-yield membership changes with horizon

Do top-yield pools stay at the top?

WEIGHTED AVERAGE

Weighted by valid date × top-k comparisons, churn rises from 13.5% after 1 day to 33.3% after 30 days.

- 1 day: 13.5% (n=2,669)
- 7 days: 23.6% (n=2,657)
- 30 days: 33.3% (n=2,612)

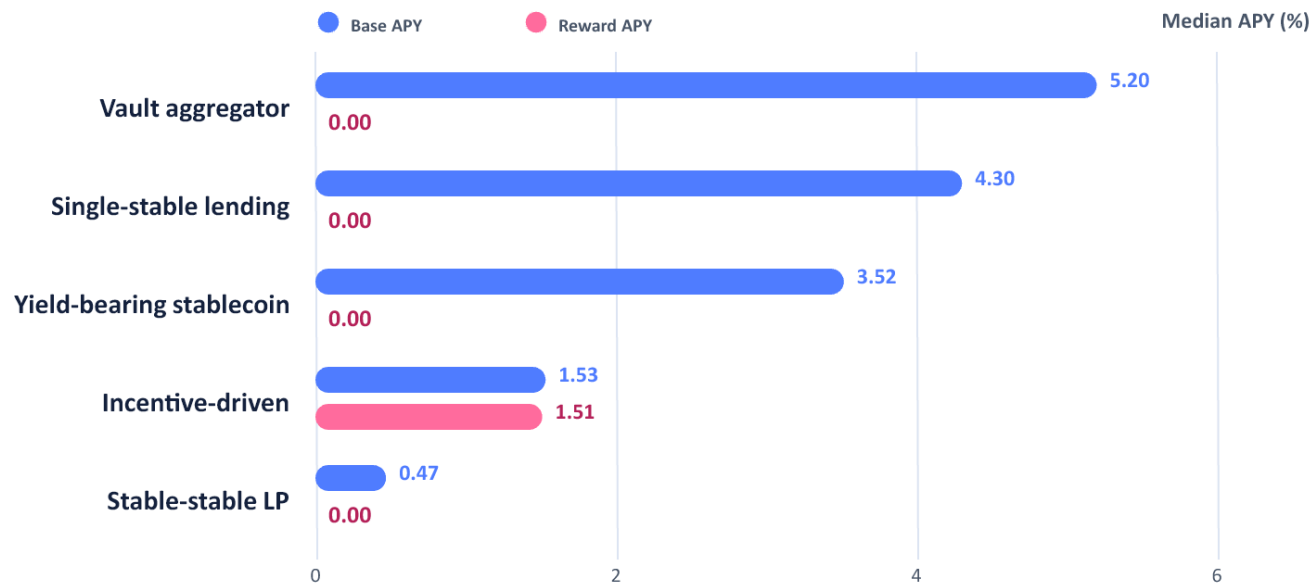


Sources: DeFiLlama, CoinGecko checks and protocol documentation. Quoted APY differs from realized return.

Mechanism changes what headline APY means

What sits behind headline APY?

Median APY components by pool type



Medians use available pool-days. Zero is observed, N/A means missing.
Component medians do not necessarily sum to median total APY.

INTERPRETATION

Components prevent false equivalence

90.3% Base APY coverage

61.6% Reward APY coverage

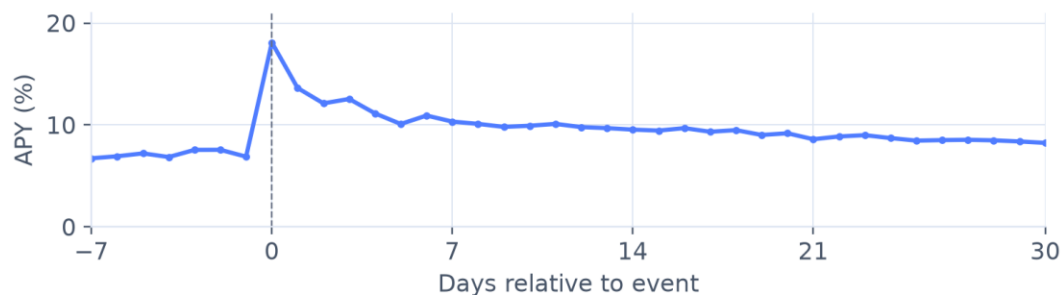
Reward coverage varies by type: 15.9% to 91.9%. Missing values are excluded.

Sources: DeFiLlama, CoinGecko and protocol documentation. Quoted APY differs from realized return.

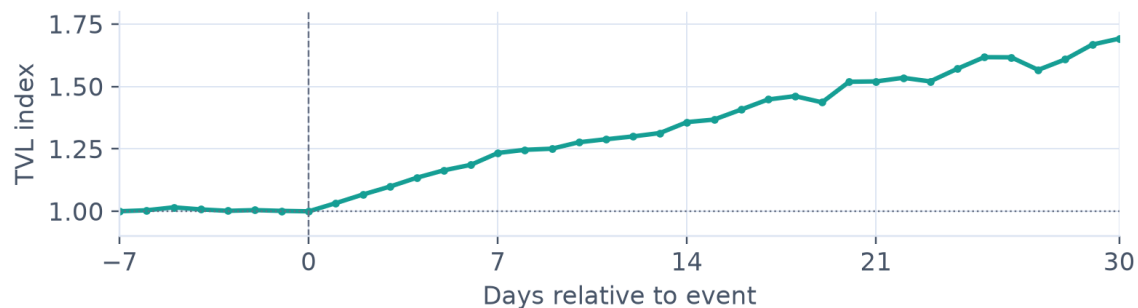
APY declines while observed TVL rises

What happens after an observed APY jump?

Median quoted APY



Median normalized TVL



OBSERVED RESULT

18.09% to 8.24%

Median APY from event day to day 30

450

Events at day 0

1.69x

Median TVL / day-0 TVL at day 30

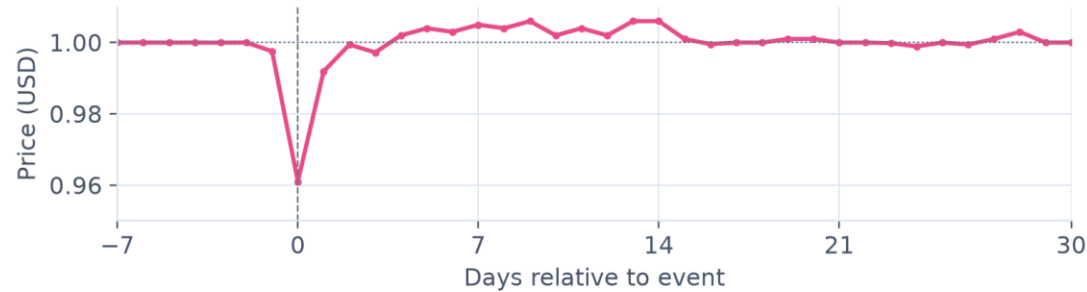
Same-event check (n=439): APY 17.96% to 8.24%, TVL 1.69x. Descriptive, non-causal.

Source: DeFiLlama. Consecutive-day jumps only. Events: 394 at -7d, 450 at 0d, 439 at +30d. Composition varies by day.

Peg stress changes the denominator behind yield

What changed during the March 2023 USDC depeg?

USDC price



Median exposed-pool APY (%)



USDC / 12 MAR 2023

\$0.9611

Minimum daily observation in this window

+1.38 pp

Median APY, day -1 to day 0

9-10

Pools per day (9 at day 0)

Changing pool composition. One descriptive case, without a control group.

Sources: DeFiLlama, CoinGecko and protocol documentation. Quoted APY differs from realized return.

Only 15.2% clear all three descriptive thresholds

Which pools combine APY, persistence and capacity?

THREE-MEDIAN SCREEN

38 of 250 pools

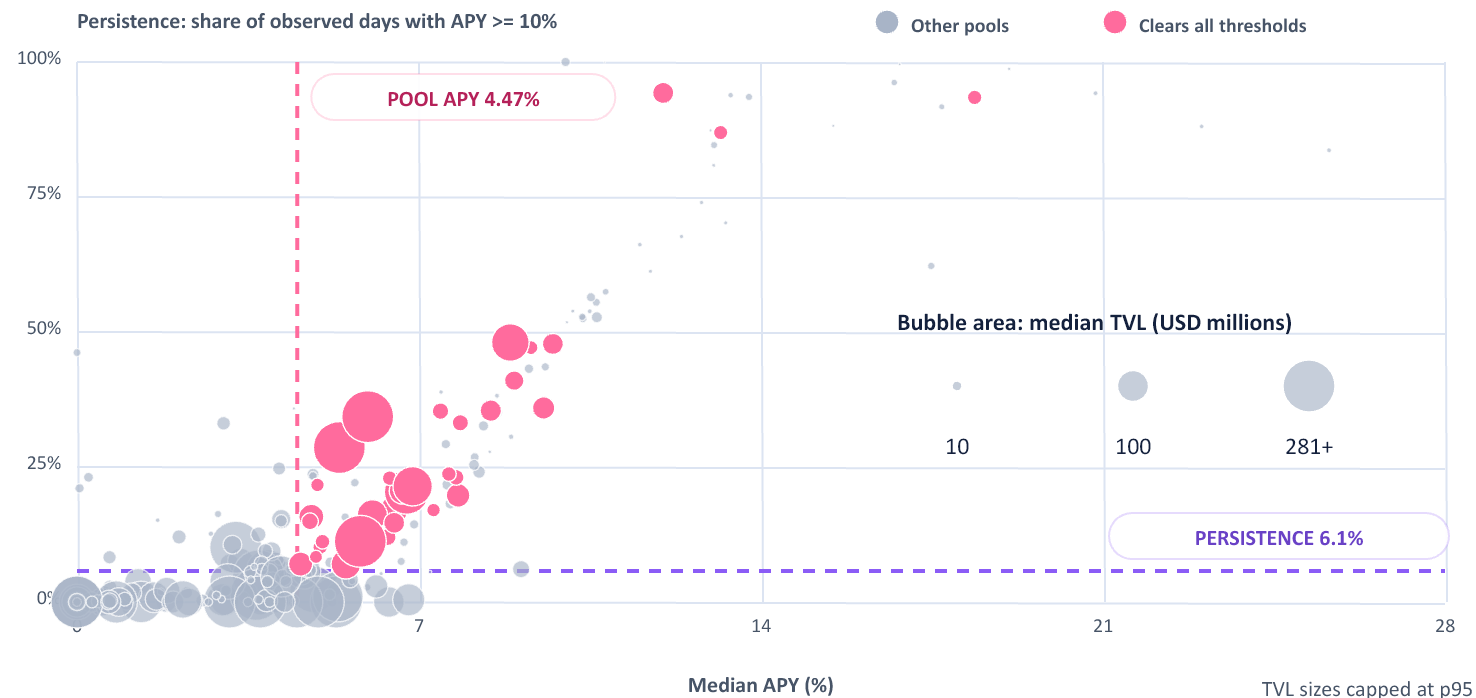
clear APY, persistence and TVL thresholds simultaneously

4.47% Median APY

6.1% Persistence

\$17.74M Median TVL

Inclusive thresholds. Bubble area encodes TVL. Descriptive screen.

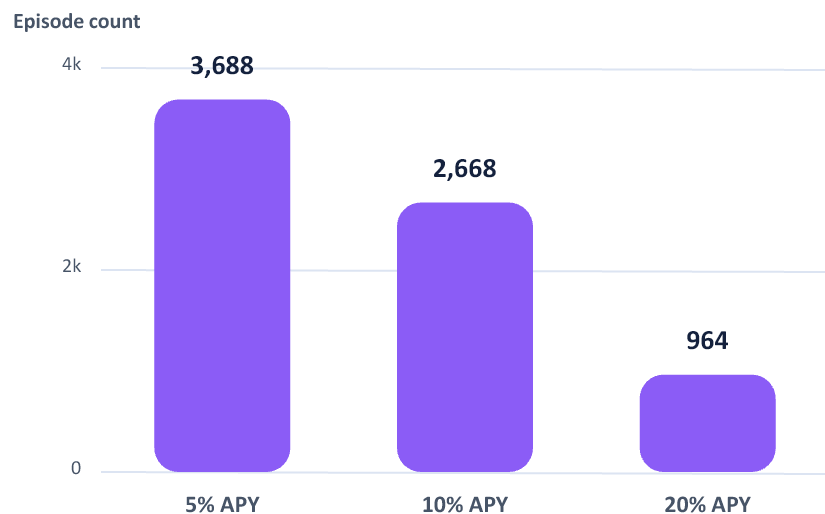


Sources: DeFiLlama, CoinGecko and protocol documentation. Quoted APY differs from realized return.

The main duration result survives alternative thresholds

Does the short-episode conclusion depend on one threshold?

Episode count changes; median duration does not



2 days

median duration at 5%,
10% and 20% APY
thresholds

Counts: 3,688 | 2,668 | 964

FIVE CHECK FAMILIES

APY threshold	5%, 10%, 20%
Minimum TVL	USD 0.5M, 1M, 5M
History length	90, 180, 365 days
Episode gap	0, 1, 3 days
Winsorization	p1 to p99

Filters change counts. Quoted APY remains distinct from realized return.

Market context is not total protocol risk

What does the project deliberately not claim?

MEASURED HERE

What the evidence supports

- Quoted APY level and persistence
- Observed TVL and reward components
- Top-k membership stability
- Stablecoin peg context
- Data quality and sensitivity checks

NOT MEASURED HERE

What the project cannot rank

- Smart-contract and oracle failure
- Counterparty and collateral quality
- Bridge, chain and governance risk
- Executable liquidity and slippage
- Investor-specific realized return

Ethical consequence: no pool ranking, no hidden safety score and no financial recommendation.

Precision has a defined scope

Day-30 survival: 6.22%

95% pointwise CI: 5.34%-7.19%. Pool bootstrap: 4.76%-7.86%.

Repeated episodes	2,000 bootstrap resamples of whole pools preserve within-pool episode dependence. The interval is wider.
Observation limits	70 episodes start at panel entry; 56 end at long gaps. Observed runs can differ from economic episodes.
Sample selection	Selection excludes short histories. The 90-day check cannot recover them. Bootstrap does not correct selection bias.
Interpretation	The evidence describes selected observed regimes. It does not estimate every pool's future yield.

Sources: episode_survival.csv and refinement_audit.json. Bootstrap: 250 pools, seed 945119, percentile interval.

Two visualization tools, complementary uses

MATPLOTLIB

Static analytical figures

This PDF uses Matplotlib for the survival curve (page 6) and daily event charts (pages 9–10).

The survival band shows uncertainty. Numerical time axes preserve the actual spacing of days.

PLOTLY

Ranking heatmap in this PDF

Page 7 contains a static export of the Plotly ranking heatmap, using the same published aggregate data.

Every cell includes its value. Reading the result requires no hover, live demo or additional file.

Analysis: Python, pandas, NumPy, lifelines, SciPy and scikit-learn

Seaborn supports static figures. ReportLab builds the report. Artifact Tool assembles the PowerPoint.

All findings are readable in this PDF. Interactive HTML and editable sources are optional supporting materials.

Code, evidence and reuse terms

github.com/Laimon99/stablecoin-yield-visualization

1. Rebuild the published evidence

```
uv sync --frozen --extra dev
uv run python scripts/refine_submission.py
```

From the cloned repository: rebuild static charts and verify event headlines from published aggregates, without API calls.

2. Recollect and rerun the method

```
uv run python scripts/reproduce_all.py --mode full
```

Live data can change. The historical raw snapshot is private. Bootstrap recalculation requires that snapshot. Deck: Artifact Tool.

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Stablecoin yield is best shown as trade-offs

The project closes on three lenses that keep a volatile number interpretable.

ONE SENTENCE DEFENSE

High stablecoin yield is common in snapshots. Persistent high yield with capacity and interpretable risk context is rarer.

PERSISTENCE

Under the main 10% definition, the median episode lasts 2 days, so duration matters as much as level.

CONTEXT

Mechanism, reward dependence and peg stress change what a nominal APY number means.

RESPONSIBLE DESIGN

The joint screen keeps APY, persistence and capacity visible without encoding a recommendation.

Code: github.com/Laimon99/stablecoin-yield-visualization
Matplotlib + Plotly. Code: MIT. Original presentation: CC BY 4.0.